

SQL Ve Pandas Database Alıřtırmaları

Yapılması İstenilenler:

Elimizde film kiralama ile ilgili bir database var. Altta 100 adet bu database ile ilgili sorular var. Bu soruların SQL sorguları ile Jupyter Notebook'da "query" aracılığı ile çözümleri var. Yapmamız gereken verilen 100 farklı soruya bunların SQL çözümlerinden yola çıkarak ve Database'i koda entegre ederek Pandas Dataframe'ine karşılık gelen kodu yazmamız tek tek. Hangi formatta yapılması gerektiği ile ilgili alta ilk 2 sorunun pandas karşılığını ekledim.

Not: Verisetimiz database(db) formatında olduğundan tabloları çekmek için SQL komutlarından yardım alabiliriz.

- Sorulara başlanmadan önce veri setinin incelenmesi ve her sorunun çözümü için öncelikle SQL sorgularının ilgili veritabanında çalıştırılması gerekir. (Bu dokümanda SQLite kullanılmış)
- Ardından sorular Pandas kütüphanesi kullanılarak Python'da yazılmalıdır. (Python için istenilen bir notebook veya IDE kullanılabilir)
- SQL çıktısı ile Python çıktısının aynı olup olmadığı karşılaştırılmalı ve dokümente edilmelidir.
- Sorgular Python ile çözülürken kesinlikle Pandas'ın **read_sql_query** fonksiyonu kullanılmamalı.
- **read_sql_query** yerine **aggregate** fonksiyonlar ile sorular çözülmeli.
- Sorgu Python'da çözülürken, **yalnızca kullanılacak tablolar** **read_sql_query** methoduyla çekilebilir.
- Aşağıda yer alan ekran fotoğraflarında sorguların hepsi tasvip **edilmeyen** yöntemeye aittir. Bu sebeple bu yöntem tercih **edilmemeli**. Kırmızı ile yazılan maddeler dikkate alınmalıdır.

Soru 1: Tüm verileri göster (Show all data)

python

 Copy code

```
import pandas as pd
import sqlite3

# Veritabanına bağlantı kur
conn = sqlite3.connect('path_to_your_database.db')

# Veritabanındaki 'rental' tablosunu pandas DataFrame olarak yükle
rental_df = pd.read_sql_query("SELECT * FROM rental", conn)

# Tüm verileri göster
print(rental_df)
```

Soru 2: Her filmde oynayan aktörleri alın (Get the actors in each movie)

python

 Copy code

```
import pandas as pd
import sqlite3

# Veritabanına bağlantı kur
conn = sqlite3.connect('path_to_your_database.db')

# Gerekli tabloları pandas DataFrame olarak yükle
film_df = pd.read_sql_query("SELECT * FROM film", conn)
film_actor_df = pd.read_sql_query("SELECT * FROM film_actor", conn)
actor_df = pd.read_sql_query("SELECT * FROM actor", conn)

# DataFrame'leri birleştir
merged_df = pd.merge(film_df, film_actor_df, on='film_id')
merged_df = pd.merge(merged_df, actor_df, on='actor_id')

# İstenen sütunları seç
result_df = merged_df[['title', 'first_name', 'last_name']]

# Her filmde oynayan aktörleri göster
print(result_df)
```

Python kursundan kurmuş olduğunuz DB:SQLİTE programından verilen database'i inceleyebilir ve ona göre gözlemleyip kodları oluşturabilirsiniz.

DB Browser for SQLite - C:\Users\Baki Akgun\OneDrive\Masaüstü\archive (2)\sqlite-sakila.db

File Edit View Tools Help

New Database Open Database Write Changes Revert Changes Open Project Save

Database Structure Browse Data Edit Pragmas Execute SQL

Table: country

	country	last_update
		Filter
1	in	2021-03-06 15:51:49
2		2021-03-06 15:51:49
3	Samoa	2021-03-06 15:51:49
4		2021-03-06 15:51:49
5		2021-03-06 15:51:49
6		2021-03-06 15:51:49
7		2021-03-06 15:51:49
8		2021-03-06 15:51:49
9		2021-03-06 15:51:49
10		2021-03-06 15:51:49
11		2021-03-06 15:51:49
12	12 Bangladesh	2021-03-06 15:51:49
13	13 Belarus	2021-03-06 15:51:49

Sorular

Soru 1: Tüm verileri göster

SQL:

sql

 Copy code

```
SELECT * FROM rental;
```

Python Pandas:

python

 Copy code

```
import pandas as pd
import sqlite3

conn = sqlite3.connect('path_to_your_database.db')
df = pd.read_sql_query("SELECT * FROM rental", conn)
print(df)
```

Soru 2: Her filmdeki oyuncularını listele

SQL:

sql

 Copy code

```
SELECT film.title, actor.first_name, actor.last_name
FROM film
JOIN film_actor ON film.film_id = film_actor.film_id
JOIN actor ON film_actor.actor_id = actor.actor_id;
```

Python Pandas:

python

 Copy code

```
query = """
SELECT film.title, actor.first_name, actor.last_name
FROM film
JOIN film_actor ON film.film_id = film_actor.film_id
JOIN actor ON film_actor.actor_id = actor.actor_id;
"""

df = pd.read_sql_query(query, conn)
print(df)
```

Soru 3: Her filmde kaç oyuncu oynadı?

SQL:

sql

 Copy code

```
SELECT film.title, COUNT(actor.actor_id) AS actor_count
FROM film
JOIN film_actor ON film.film_id = film_actor.film_id
JOIN actor ON film_actor.actor_id = actor.actor_id
GROUP BY film.title;
```

Python Pandas:

python

 Copy code

```
query = """
SELECT film.title, COUNT(actor.actor_id) AS actor_count
FROM film
JOIN film_actor ON film.film_id = film_actor.film_id
JOIN actor ON film_actor.actor_id = actor.actor_id
GROUP BY film.title;
"""

df = pd.read_sql_query(query, conn)
print(df)
```

Soru 4: Her oyuncu kaç filmde oynadı?

SQL:

sql

 Copy code

```
SELECT actor.first_name, actor.last_name, COUNT(film.film_id) AS movie_count
FROM actor
JOIN film_actor ON actor.actor_id = film_actor.actor_id
JOIN film ON film_actor.film_id = film.film_id
GROUP BY actor.first_name, actor.last_name;
```

Python Pandas:

python

 Copy code

```
query = """
SELECT actor.first_name, actor.last_name, COUNT(film.film_id) AS movie_count
FROM actor
JOIN film_actor ON actor.actor_id = film_actor.actor_id
JOIN film ON film_actor.film_id = film.film_id
GROUP BY actor.first_name, actor.last_name;
"""

df = pd.read_sql_query(query, conn)
print(df)
```

Soru 5: Envanterde olmayan filmler var mı ve varsa kaç tane?

```
SQL:

sql
Copy code

SELECT COUNT(*) AS not_in_inventory
FROM film
WHERE film_id NOT IN (SELECT film_id FROM inventory);

Python Pandas:

python
Copy code

query = """
SELECT COUNT(*) AS not_in_inventory
FROM film
WHERE film_id NOT IN (SELECT film_id FROM inventory);
"""

df = pd.read_sql_query(query, conn)
print(df)
```

Soru 6: Kiralanabilir olan her filmin kaç kez kiralandığını ve toplam gelirlerini getirin

```
SQL:

sql
Copy code

SELECT film.title, COUNT(rental.rental_id) AS rental_count, SUM(payment.amount) AS total_r
FROM film
JOIN inventory ON film.film_id = inventory.film_id
JOIN rental ON inventory.inventory_id = rental.inventory_id
JOIN payment ON rental.rental_id = payment.rental_id
GROUP BY film.title;

Python Pandas:

python
Copy code

query = """
SELECT film.title, COUNT(rental.rental_id) AS rental_count, SUM(payment.amount) AS total_r
FROM film
JOIN inventory ON film.film_id = inventory.film_id
JOIN rental ON inventory.inventory_id = rental.inventory_id
JOIN payment ON rental.rental_id = payment.rental_id
GROUP BY film.title;
"""

df = pd.read_sql_query(query, conn)
print(df)
```

Soru 7: Envanterde olmayan filmlerin kira oranlarını getirin

SQL:

sql

Copy code

```
SELECT title, rental_rate
FROM film
WHERE film_id NOT IN (SELECT film_id FROM inventory);
```

Python Pandas:

python

Copy code

```
query = """
SELECT title, rental_rate
FROM film
WHERE film_id NOT IN (SELECT film_id FROM inventory);
"""

df = pd.read_sql_query(query, conn)
print(df)
```

Soru 8: Birden fazla DVD'yi iade etmeyen kaç müşteri var?

SQL:

sql

Copy code

```
SELECT COUNT(DISTINCT customer_id) AS customer_count
FROM rental
WHERE return_date IS NULL
GROUP BY customer_id
HAVING COUNT(rental_id) > 1;
```

Python Pandas:

python

Copy code

```
query = """
SELECT COUNT(DISTINCT customer_id) AS customer_count
FROM rental
WHERE return_date IS NULL
GROUP BY customer_id
HAVING COUNT(rental_id) > 1;
"""

df = pd.read_sql_query(query, conn)
print(df)
```

Soru 9: Her müşteri kaç film kiraladı?

SQL:

sql

 Copy code

```
SELECT customer.first_name, customer.last_name, COUNT(rental.rental_id) AS rental_count
FROM customer
JOIN rental ON customer.customer_id = rental.customer_id
GROUP BY customer.first_name, customer.last_name;
```

Python Pandas:

python

 Copy code

```
query = """
SELECT customer.first_name, customer.last_name, COUNT(rental.rental_id) AS rental_count
FROM customer
JOIN rental ON customer.customer_id = rental.customer_id
GROUP BY customer.first_name, customer.last_name;
"""

df = pd.read_sql_query(query, conn)
print(df)
```

Soru 10: Türlerine göre en çok kiralanan filmler ve bunlara ne kadar ödendi?

SQL:

sql

Copy code

```
SELECT film.title, COUNT(rental.rental_id) AS rental_count, SUM(payment.amount) AS total_revenue
FROM category
JOIN film_category ON category.category_id = film_category.category_id
JOIN film ON film_category.film_id = film.film_id
JOIN inventory ON film.film_id = inventory.film_id
JOIN rental ON inventory.inventory_id = rental.inventory_id
JOIN payment ON rental.rental_id = payment.rental_id
GROUP BY film.title
```

Python Pandas:

python

Copy code

```
query = """
SELECT category.name AS genre, film.title, COUNT(rental.rental_id) AS rental_count, SUM(payment.amount) AS total_revenue
FROM category
JOIN film_category ON category.category_id = film_category.category_id
JOIN film ON film_category.film_id = film.film_id
JOIN inventory ON film.film_id = inventory.film_id
JOIN rental ON inventory.inventory_id = rental.inventory_id
JOIN payment ON rental.rental_id = payment.rental_id
GROUP BY category.name, film.title
ORDER BY rental_count DESC;
"""

df = pd.read_sql_query(query, conn)
print(df)
```

Soru 11: Tür ve Tarihe Göre Kiralama Sayısı ve Gelir

SQL:

sql

Copy code

```
SELECT category.name AS genre, DATE(rental.rental_date) AS rental_date, COUNT(rental.rental_id) AS rental_count
FROM category
JOIN film_category ON category.category_id = film_category.category_id
JOIN film ON film_category.film_id = film.film_id
JOIN inventory ON film.film_id = inventory.film_id
JOIN rental ON inventory.inventory_id = rental.inventory_id
JOIN payment ON rental.rental_id = payment.rental_id
GROUP BY genre, rental_date
ORDER BY rental_date;
```

Python Pandas:

python

Copy code

```
query = """
SELECT category.name AS genre, DATE(rental.rental_date) AS rental_date, COUNT(rental.rental_id) AS rental_count
FROM category
JOIN film_category ON category.category_id = film_category.category_id
JOIN film ON film_category.film_id = film.film_id
JOIN inventory ON film.film_id = inventory.film_id
JOIN rental ON inventory.inventory_id = rental.inventory_id
JOIN payment ON rental.rental_id = payment.rental_id
GROUP BY genre, rental_date
ORDER BY rental_date;
"""

df = pd.read_sql_query(query, conn)
print(df)
```

Soru 12: Kiralanabilir Filmler İçin Türlerine Göre Her Filmin Kaç Kez Kiralandığı

SQL:

sql

 Copy code

```
SELECT category.name AS genre, film.title, COUNT(rental.rental_id) AS rental_count
FROM category
JOIN film_category ON category.category_id = film_category.category_id
JOIN film ON film_category.film_id = film.film_id
JOIN inventory ON film.film_id = inventory.film_id
JOIN rental ON inventory.inventory_id = rental.inventory_id
GROUP BY genre, film.title;
```

Python Pandas:

python

 Copy code

```
query = """
SELECT category.name AS genre, film.title, COUNT(rental.rental_id) AS rental_count
FROM category
JOIN film_category ON category.category_id = film_category.category_id
JOIN film ON film_category.film_id = film.film_id
JOIN inventory ON film.film_id = inventory.film_id
JOIN rental ON inventory.inventory_id = rental.inventory_id
GROUP BY genre, film.title;
"""

df = pd.read_sql_query(query, conn)
print(df)
```

Soru 13: En çok rafta bekleyen filmler

sql

 Copy code

```
SELECT film.title, DATEDIFF(NOW(), rental_date) AS days_on_shelf
FROM film
JOIN inventory ON film.film_id = inventory.film_id
JOIN rental ON inventory.inventory_id = rental.inventory_id
WHERE rental.return_date IS NULL
ORDER BY days_on_shelf DESC;
```

Python Pandas:

python

 Copy code

```
query = """
SELECT film.title, DATE('now') - rental_date AS days_on_shelf
FROM film
JOIN inventory ON film.film_id = inventory.film_id
JOIN rental ON inventory.inventory_id = rental.inventory_id
WHERE rental.return_date IS NULL
ORDER BY days_on_shelf DESC;
"""

df = pd.read_sql_query(query, conn)
print(df)
```

Soru 14: Geç, Erken ve Zamanında İade Edilen Kiralanmış Filmler

```
sql                                                                    Copy code

SELECT
    SUM(CASE WHEN return_date > due_date THEN 1 ELSE 0 END) AS late_returns,
    SUM(CASE WHEN return_date < due_date THEN 1 ELSE 0 END) AS early_returns,
    SUM(CASE WHEN return_date = due_date THEN 1 ELSE 0 END) AS on_time_returns
FROM rental;
```

Python Pandas:

```
python                                                                    Copy code

query = """
SELECT
    SUM(CASE WHEN return_date > due_date THEN 1 ELSE 0 END) AS late_returns,
    SUM(CASE WHEN return_date < due_date THEN 1 ELSE 0 END) AS early_returns,
    SUM(CASE WHEN return_date = due_date THEN 1 ELSE 0 END) AS on_time_returns
FROM rental;
"""

df = pd.read_sql_query(query, conn)
print(df)
```

Soru 15: Hangi müşteri en çok DVD kiralamış?

```
sql                                                                    Copy code
er.first_name, customer.last_name, COUNT(rental.rental_id) AS rental_count

| customer.customer_id = rental.customer_id
mer.first_name, customer.last_name
l_count DESC

Python Pandas:

python                                                                    Copy code

query = """
SELECT customer.first_name, customer.last_name, COUNT(rental.rental_id) AS
FROM customer
JOIN rental ON customer.customer_id = rental.customer_id
GROUP BY customer.first_name, customer.last_name
ORDER BY rental_count DESC
LIMIT 1;
"""

df = pd.read_sql_query(query, conn)
print(df)
```

Soru 16: En popüler film kategorisi nedir?

sql

Copy code

```
SELECT category.name, COUNT(rental.rental_id) AS rental_count
FROM category
JOIN film_category ON category.category_id = film_category.category_id
JOIN film ON film_category.film_id = film.film_id
JOIN inventory ON film.film_id = inventory.film_id
JOIN rental ON inventory.inventory_id = rental.inventory_id
GROUP BY category.name
ORDER BY rental_count DESC
LIMIT 1;
```

Python Pandas:

python

Copy code

```
query = """
SELECT category.name, COUNT(rental.rental_id) AS rental_count
FROM category
JOIN film_category ON category.category_id = film_category.category_id
JOIN film ON film_category.film_id = film.film_id
JOIN inventory ON film.film_id = inventory.film_id
JOIN rental ON inventory.inventory_id = rental.inventory_id
GROUP BY category.name
ORDER BY rental_count DESC
LIMIT 1;
"""

df = pd.read_sql_query(query, conn)
print(df)
```

Soru 17: Hangi çalışan en çok kiralama işlemi gerçekleştirmiş?

SQL:

sql

Copy code

```
staff.first_name, staff.last_name, COUNT(rental.rental_id) AS rental_count  
FROM staff  
JOIN rental ON staff.staff_id = rental.staff_id  
GROUP BY staff.first_name, staff.last_name  
ORDER BY rental_count DESC  
LIMIT 1;
```

Python Pandas:

python

Copy code

```
query = """  
SELECT staff.first_name, staff.last_name, COUNT(rental.rental_id) AS rental_count  
FROM staff  
JOIN rental ON staff.staff_id = rental.staff_id  
GROUP BY staff.first_name, staff.last_name  
ORDER BY rental_count DESC  
LIMIT 1;  
"""  
  
df = pd.read_sql_query(query, conn)  
print(df)
```

Soru 18: En çok geliri hangi film getirmiş?

sql

 Copy code

```
SELECT film.title, SUM(payment.amount) AS total_revenue
FROM film
JOIN inventory ON film.film_id = inventory.film_id
JOIN rental ON inventory.inventory_id = rental.inventory_id
JOIN payment ON rental.rental_id = payment.rental_id
GROUP BY film.title
ORDER BY total_revenue DESC
LIMIT 1;
```

Python Pandas:

python

 Copy code

```
query = """
SELECT film.title, SUM(payment.amount) AS total_revenue
FROM film
JOIN inventory ON film.film_id = inventory.film_id
JOIN rental ON inventory.inventory_id = rental.inventory_id
JOIN payment ON rental.rental_id = payment.rental_id
GROUP BY film.title
ORDER BY total_revenue DESC
LIMIT 1;
"""

df = pd.read_sql_query(query, conn)
print(df)
```



Soru 19: Her müşteri için toplam harcama miktarını bulun

SQL:

sql

Copy code

```
customer.first_name, customer.last_name, SUM(payment.amount) AS total_spent
FROM customer
JOIN rental ON customer.customer_id = rental.customer_id
JOIN payment ON rental.rental_id = payment.rental_id
GROUP BY customer.first_name, customer.last_name;
```

Python Pandas:

python

Copy code

```
query = """
SELECT customer.first_name, customer.last_name, SUM(payment.amount) AS total_spent
FROM customer
JOIN rental ON customer.customer_id = rental.customer_id
JOIN payment ON rental.rental_id = payment.rental_id
GROUP BY customer.first_name, customer.last_name;
"""

df = pd.read_sql_query(query, conn)
print(df)
```

Soru 20: Her kategorideki toplam kiralama sayısını ve gelirleri bulun

```
sql                                                                    Copy code

COUNT(rental.rental_id) AS rental_count, SUM(payment.amount) AS total_revenue

category.category_id = film_category.category_id
category.film_id = film.film_id
inventory.film_id = inventory.film_id
inventory.inventory_id = rental.inventory_id
payment.rental_id = payment.rental_id

Python Pandas:

python                                                                    Copy code

query = """
SELECT category.name, COUNT(rental.rental_id) AS rental_count, SUM(payment
FROM category
JOIN film_category ON category.category_id = film_category.category_id
JOIN film ON film_category.film_id = film.film_id
JOIN inventory ON film.film_id = inventory.film_id
JOIN rental ON inventory.inventory_id = rental.inventory_id
JOIN payment ON rental.rental_id = payment.rental_id
GROUP BY category.name;
"""

df = pd.read_sql_query(query, conn)
print(df)
```

Soru 21: En uzun süre kirada kalmış filmleri bulun

```
SELECT film.title, MAX(DATEDIFF(return_date, rental_date)) AS max_rental_duration
FROM film
JOIN inventory ON film.film_id = inventory.film_id
JOIN rental ON inventory.inventory_id = rental.inventory_id
WHERE rental.return_date IS NOT NULL
GROUP BY film.title
ORDER BY max_rental_duration DESC
LIMIT 10;
```

Python Pandas:

python

Copy code

```
query = """
SELECT film.title, MAX(return_date - rental_date) AS max_rental_duration
FROM film
JOIN inventory ON film.film_id = inventory.film_id
JOIN rental ON inventory.inventory_id = rental.inventory_id
WHERE rental.return_date IS NOT NULL
GROUP BY film.title
ORDER BY max_rental_duration DESC
LIMIT 10;
"""

df = pd.read_sql_query(query, conn)
print(df)
```

Continue generating

Soru 22: En az kiralanan 5 film hangileridir?

sql

Copy code

```
SELECT film.title, COUNT(rental.rental_id) AS rental_count
FROM film
JOIN inventory ON film.film_id = inventory.film_id
JOIN rental ON inventory.inventory_id = rental.inventory_id
GROUP BY film.title
ORDER BY rental_count ASC
LIMIT 5;
```

Python Pandas:

python

Copy code

```
query = """
SELECT film.title, COUNT(rental.rental_id) AS rental_count
FROM film
JOIN inventory ON film.film_id = inventory.film_id
JOIN rental ON inventory.inventory_id = rental.inventory_id
GROUP BY film.title
ORDER BY rental_count ASC
LIMIT 5;
"""

df = pd.read_sql_query(query, conn)
print(df)
```

Soru 24: En fazla kazanç sağlayan 5 müşteriyi bulun

```
sql                                                                    Copy code

customer.first_name, customer.last_name, SUM(payment.amount) AS total_spent
FROM customer
JOIN rental ON customer.customer_id = rental.customer_id
JOIN payment ON rental.rental_id = payment.rental_id
GROUP BY customer.first_name, customer.last_name
ORDER BY total_spent DESC

Python Pandas:

python                                                                    Copy code

query = """
SELECT customer.first_name, customer.last_name, SUM(payment.amount) AS total_spent
FROM customer
JOIN rental ON customer.customer_id = rental.customer_id
JOIN payment ON rental.rental_id = payment.rental_id
GROUP BY customer.first_name, customer.last_name
ORDER BY total_spent DESC
LIMIT 5;
"""

df = pd.read_sql_query(query, conn)
print(df)
```

Soru 25: Her filmin ortalama kiralanma süresini bulun

```
sql                                                                    Copy code

film.title, AVG(DATEDIFF(return_date, rental_date)) AS avg_rental_duration
.m
rental ON film.film_id = inventory.film_id
rental ON inventory.inventory_id = rental.inventory_id
rental.return_date IS NOT NULL
/ film.title;
```

Python Pandas:

```
python                                                                    Copy code

query = """
SELECT film.title, AVG(return_date - rental_date) AS avg_rental_duration
FROM film
JOIN inventory ON film.film_id = inventory.film_id
JOIN rental ON inventory.inventory_id = rental.inventory_id
WHERE rental.return_date IS NOT NULL
GROUP BY film.title;
"""

df = pd.read_sql_query(query, conn)
print(df)
```

Soru 26: Her türde en popüler filmi bulun

```
sql                                                                    Copy code

category.name AS genre, film.title, COUNT(rental.rental_id) AS rental_count
FROM category
JOIN film_category ON category.category_id = film_category.category_id
JOIN film ON film_category.film_id = film.film_id
JOIN inventory ON film.film_id = inventory.film_id
JOIN rental ON inventory.inventory_id = rental.inventory_id
GROUP BY genre, film.title
ORDER BY rental_count DESC
LIMIT 1;
```

Python Pandas:

```
python                                                                    Copy code

query = """
SELECT category.name AS genre, film.title, COUNT(rental.rental_id) AS rental_count
FROM category
JOIN film_category ON category.category_id = film_category.category_id
JOIN film ON film_category.film_id = film.film_id
JOIN inventory ON film.film_id = inventory.film_id
JOIN rental ON inventory.inventory_id = rental.inventory_id
GROUP BY genre, film.title
ORDER BY rental_count DESC
LIMIT 1;
"""

df = pd.read_sql_query(query, conn)
print(df)
```

Soru 27: Her türde en fazla gelir sağlayan filmi bulun

```
SELECT category.name AS genre, film.title, SUM(payment.amount) AS total_revenue
FROM category
JOIN film_category ON category.category_id = film_category.category_id
JOIN film ON film_category.film_id = film.film_id
JOIN inventory ON film.film_id = inventory.film_id
JOIN rental ON inventory.inventory_id = rental.inventory_id
JOIN payment ON rental.rental_id = payment.rental_id
GROUP BY genre, film.title
ORDER BY total_revenue DESC
LIMIT 1;
```

Python Pandas:

python

 Copy code

```
query = """
SELECT category.name AS genre, film.title, SUM(payment.amount) AS total_r
FROM category
JOIN film_category ON category.category_id = film_category.category_id
JOIN film ON film_category.film_id = film.film_id
JOIN inventory ON film.film_id = inventory.film_id
JOIN rental ON inventory.inventory_id = rental.inventory_id
JOIN payment ON rental.rental_id = payment.rental_id
GROUP BY genre, film.title
ORDER BY total_revenue DESC
LIMIT 1;
"""

df = pd.read_sql_query(query, conn)
print(df)
```



Soru 28: En çok DVD iade etmeyen müşteriyi bulun

```
sql                                                                    Copy code

first_name, customer.last_name, COUNT(rental.rental_id) AS unreturned_count

customer.customer_id = rental.customer_id
return_date IS NULL
r.first_name, customer.last_name
unreturned_count DESC

Python Pandas:

python                                                                    Copy code

query = """
SELECT customer.first_name, customer.last_name, COUNT(rental.rental_id) AS
FROM customer
JOIN rental ON customer.customer_id = rental.customer_id
WHERE rental.return_date IS NULL
GROUP BY customer.first_name, customer.last_name
ORDER BY unreturned_count DESC
LIMIT 1;
"""

df = pd.read_sql_query(query, conn)
print(df)
```

Soru 29: En fazla kiralama yapan 5 çalışanı bulun

```
sql                                                                    Copy code

staff.first_name, staff.last_name, COUNT(rental.rental_id) AS rental_count
FROM staff
JOIN rental ON staff.staff_id = rental.staff_id
GROUP BY staff.first_name, staff.last_name
ORDER BY rental_count DESC
LIMIT 5;
```

Python Pandas:

```
python                                                                    Copy code

query = """
SELECT staff.first_name, staff.last_name, COUNT(rental.rental_id) AS rental_count
FROM staff
JOIN rental ON staff.staff_id = rental.staff_id
GROUP BY staff.first_name, staff.last_name
ORDER BY rental_count DESC
LIMIT 5;
"""

df = pd.read_sql_query(query, conn)
print(df)
```

Soru 30: En fazla kiralama yapan 5 müşteri hangi şubeden kiralama yapmış?

```
sql                                                                    Copy code

er.first_name, customer.last_name, COUNT(rental.rental_id) AS rental_count

er_id = rental.customer_id
tory_id = inventory.inventory_id
_id = store.store_id
omer.first_name, customer.last_name

Python Pandas:

python                                                                    Copy code

query = """
SELECT store.store_id, customer.first_name, customer.last_name, COUNT(rental.rental_id) AS rental_count
FROM customer
JOIN rental ON customer.customer_id = rental.customer_id
JOIN inventory ON rental.inventory_id = inventory.inventory_id
JOIN store ON inventory.store_id = store.store_id
GROUP BY store.store_id, customer.first_name, customer.last_name
ORDER BY rental_count DESC
LIMIT 5;
"""

df = pd.read_sql_query(query, conn)
print(df)
```

Soru 31: Her türde en az kiralanan filmi bulun

```
sql                                                                    Copy code

category.name AS genre, film.title, COUNT(rental.rental_id) AS rental_count
category
n_category ON category.category_id = film_category.category_id
n ON film_category.film_id = film.film_id
entory ON film.film_id = inventory.film_id
cal ON inventory.inventory_id = rental.inventory_id
genre, film.title
rental_count ASC

Python Pandas:

python                                                                    Copy code

query = """
SELECT category.name AS genre, film.title, COUNT(rental.rental_id) AS ren
FROM category
JOIN film_category ON category.category_id = film_category.category_id
JOIN film ON film_category.film_id = film.film_id
JOIN inventory ON film.film_id = inventory.film_id
JOIN rental ON inventory.inventory_id = rental.inventory_id
GROUP BY genre, film.title
ORDER BY rental_count ASC
LIMIT 1;
"""

df = pd.read_sql_query(query, conn)
print(df)
```

Soru 32: En çok kiralama yapan 5 müşteri hangi şehirde?

```
sql                                                                    Copy code

er.first_name, customer.last_name, COUNT(rental.rental_id) AS rental_count

address_id = address.address_id
city_id = city.city_id
customer_id = rental.customer_id
customer.first_name, customer.last_name
SC

Python Pandas:

python                                                                    Copy code

query = """
SELECT city.city, customer.first_name, customer.last_name, COUNT(rental.rental_id) AS rental_count
FROM customer
JOIN address ON customer.address_id = address.address_id
JOIN city ON address.city_id = city.city_id
JOIN rental ON customer.customer_id = rental.customer_id
GROUP BY city.city, customer.first_name, customer.last_name
ORDER BY rental_count DESC
LIMIT 5;
"""

df = pd.read_sql_query(query, connection)
print(df)
```

Soru 33: En çok kazanç sağlayan 5 müşteriyi hangi şehirde bulun?

```
sql                                                                    Copy code

customer.first_name, customer.last_name, SUM(payment.amount) AS total_spent

customer.address_id = address.address_id
customer.city_id = city.city_id
customer.customer_id = rental.customer_id
rental.rental_id = payment.rental_id
customer.first_name, customer.last_name
: DESC

Python Pandas:

python                                                                    Copy code

query = """
SELECT city.city, customer.first_name, customer.last_name, SUM(payment.a
FROM customer
JOIN address ON customer.address_id = address.address_id
JOIN city ON address.city_id = city.city_id
JOIN rental ON customer.customer_id = rental.customer_id
JOIN payment ON rental.rental_id = payment.rental_id
GROUP BY city.city, customer.first_name, customer.last_name
ORDER BY total_spent DESC
LIMIT 5;
"""

df = pd.read_sql_query(query, conn)
print(df)
```

Soru 34: En çok kiralanan 5 filmi hangi şehirde bulun?

```
SELECT city.city, film.title, COUNT(rental.rental_id) AS rental_count
FROM city
JOIN address ON city.city_id = address.city_id
JOIN customer ON address.address_id = customer.address_id
JOIN rental ON customer.customer_id = rental.customer_id
JOIN inventory ON rental.inventory_id = inventory.inventory_id
JOIN film ON inventory.film_id = film.film_id
GROUP BY city.city, film.title
ORDER BY rental_count DESC
LIMIT 5;
```

Python Pandas:

python

 Copy code

```
query = """
SELECT city.city, film.title, COUNT(rental.rental_id) AS rental_count
FROM city
JOIN address ON city.city_id = address.city_id
JOIN customer ON address.address_id = customer.address_id
JOIN rental ON customer.customer_id = rental.customer_id
JOIN inventory ON rental.inventory_id = inventory.inventory_id
JOIN film ON inventory.film_id = film.film_id
GROUP BY city.city, film.title
ORDER BY rental_count DESC
LIMIT 5;
"""

df = pd.read_sql_query(query, conn)
print(df)
```

Soru 35: En az kiralanan 5 filmi hangi şehirde bulun?

```
SELECT city.city, film.title, COUNT(rental.rental_id) AS rental_count
FROM city
JOIN address ON city.city_id = address.city_id
JOIN customer ON address.address_id = customer.address_id
JOIN rental ON customer.customer_id = rental.customer_id
JOIN inventory ON rental.inventory_id = inventory.inventory_id
JOIN film ON inventory.film_id = film.film_id
GROUP BY city.city, film.title
ORDER BY rental_count ASC
LIMIT 5;
```

Python Pandas:

python

 Copy code

```
query = """
SELECT city.city, film.title, COUNT(rental.rental_id) AS rental_count
FROM city
JOIN address ON city.city_id = address.city_id
JOIN customer ON address.address_id = customer.address_id
JOIN rental ON customer.customer_id = rental.customer_id
JOIN inventory ON rental.inventory_id = inventory.inventory_id
JOIN film ON inventory.film_id = film.film_id
GROUP BY city.city, film.title
ORDER BY rental_count ASC
LIMIT 5;
"""

df = pd.read_sql_query(query, conn)
print(df)
```

Soru 36: En çok kazanç sağlayan 5 filmi hangi şehirde bulun?

```
SELECT city.city, film.title, SUM(payment.amount) AS total_revenue
FROM city
JOIN address ON city.city_id = address.city_id
JOIN customer ON address.address_id = customer.address_id
JOIN rental ON customer.customer_id = rental.customer_id
JOIN inventory ON rental.inventory_id = inventory.inventory_id
JOIN film ON inventory.film_id = film.film_id
JOIN payment ON rental.rental_id = payment.rental_id
GROUP BY city.city, film.title
ORDER BY total_revenue DESC
LIMIT 5;
```

Python Pandas:

python

 Copy code

```
query = """
SELECT city.city, film.title, SUM(payment.amount) AS total_revenue
FROM city
JOIN address ON city.city_id = address.city_id
JOIN customer ON address.address_id = customer.address_id
JOIN rental ON customer.customer_id = rental.customer_id
JOIN inventory ON rental.inventory_id = inventory.inventory_id
JOIN film ON inventory.film_id = film.film_id
JOIN payment ON rental.rental_id = payment.rental_id
GROUP BY city.city, film.title
ORDER BY total_revenue DESC
LIMIT 5;
"""

df = pd.read_sql_query(query, conn)
print(df)
```



Soru 37: En az kazanç sağlayan 5 filmi hangi şehirde bulun?

```
SELECT city.city, film.title, SUM(payment.amount) AS total_revenue
FROM city
JOIN address ON city.city_id = address.city_id
JOIN customer ON address.address_id = customer.address_id
JOIN rental ON customer.customer_id = rental.customer_id
JOIN inventory ON rental.inventory_id = inventory.inventory_id
JOIN film ON inventory.film_id = film.film_id
JOIN payment ON rental.rental_id = payment.rental_id
GROUP BY city.city, film.title
ORDER BY total_revenue ASC
LIMIT 5;
```

Python Pandas:

python

 Copy code

```
query = """
SELECT city.city, film.title, SUM(payment.amount) AS total_revenue
FROM city
JOIN address ON city.city_id = address.city_id
JOIN customer ON address.address_id = customer.address_id
JOIN rental ON customer.customer_id = rental.customer_id
JOIN inventory ON rental.inventory_id = inventory.inventory_id
JOIN film ON inventory.film_id = film.film_id
JOIN payment ON rental.rental_id = payment.rental_id
GROUP BY city.city, film.title
ORDER BY total_revenue ASC
LIMIT 5;
"""

df = pd.read_sql_query(query, conn)
print(df)
```



Soru 38: En fazla kiralama yapan müşteri hangi filmleri kiralamış?

```
sql Copy code

SELECT customer.first_name, customer.last_name, category.name AS genre, COUNT(rental.rental_id) AS total_rentals
FROM customer
JOIN rental ON customer.customer_id = rental.customer_id
JOIN inventory ON rental.inventory_id = inventory.inventory_id
JOIN film ON inventory.film_id = film.film_id
JOIN film_category ON film.film_id = film_category.film_id
JOIN category ON film_category.category_id = category.category_id
WHERE customer.customer_id = (
    SELECT customer.customer_id
    FROM customer
    JOIN rental ON customer.customer_id = rental.customer_id
    GROUP BY customer.customer_id
    ORDER BY COUNT(rental.rental_id) DESC
    LIMIT 1
)
GROUP BY customer.first_name, customer.last_name, genre;
```

python Pandas:

```
python Copy code

query = """
SELECT customer.first_name, customer.last_name, category.name AS genre, COUNT(rental.rental_id) AS total_rentals
FROM customer
JOIN rental ON customer.customer_id = rental.customer_id
JOIN inventory ON rental.inventory_id = inventory.inventory_id
JOIN film ON inventory.film_id = film.film_id
JOIN film_category ON film.film_id = film_category.film_id
JOIN category ON film_category.category_id = category.category_id
WHERE customer.customer_id = (
    SELECT customer.customer_id
    FROM customer
    JOIN rental ON customer.customer_id = rental.customer_id
    GROUP BY customer.customer_id
```

Soru 40: En çok kazanç sağlayan müşteri hangi filmleri kiralamış?

```
sql Copy code

SELECT customer.first_name, customer.last_name, film.title, SUM(payment.amount) AS total_sp
FROM customer
JOIN rental ON customer.customer_id = rental.customer_id
JOIN inventory ON rental.inventory_id = inventory.inventory_id
JOIN film ON inventory.film_id = film.film_id
JOIN payment ON rental.rental_id = payment.rental_id
WHERE customer.customer_id = (
    SELECT customer.customer_id
    FROM customer
    JOIN rental ON customer.customer_id = rental.customer_id
    JOIN payment ON rental.rental_id = payment.rental_id
    GROUP BY customer.customer_id
    ORDER BY SUM(payment.amount) DESC
    LIMIT 1
)
GROUP BY customer.first_name, customer.last_name,
```

Soru 41: En az kazanç sağlayan müşteri hangi filmleri kiralamış?

```
sql Copy code

CT customer.first_name, customer.last_name, film.title, SUM(payment.amount) AS total_spent
customer
rental ON customer.customer_id = rental.customer_id
inventory ON rental.inventory_id = inventory.inventory_id
film ON inventory.film_id = film.film_id
payment ON rental.rental_id = payment.rental_id
E customer.customer_id = (
SELECT customer.customer_id
FROM customer
JOIN rental ON customer.customer_id = rental.customer_id
JOIN payment ON rental.rental_id = payment.rental_id
GROUP BY customer.customer_id
ORDER BY SUM(payment.amount) ASC
LIMIT 1

P BY customer.first_name, customer.last_name, film.title;
```

Python Pandas:

```
python Copy code

query = """
SELECT customer.first_name, customer.last_name, film.title, SUM(payment.amount) AS total_s
FROM customer
JOIN rental ON customer.customer_id = rental.customer_id
JOIN inventory ON rental.inventory_id = inventory.inventory_id
JOIN film ON inventory.film_id = film.film_id
JOIN payment ON rental.rental_id = payment.rental_id
WHERE customer.customer_id = (
    SELECT customer.customer_id
    FROM customer
    JOIN rental ON customer.customer_id = rental.customer_id
    JOIN payment ON rental.rental_id = payment.rental_id
    GROUP BY customer.customer_id
    ORDER BY SUM(payment.amount) ASC
    LIMIT 1
)
GROUP BY customer.first_name, customer.last_name, film.title;
"""

df = pd.read_sql_query(query, conn)
print(df)
```

Soru 42: En az kazanç sağlayan müşteri hangi türde en fazla film kiralamış?

```
sql                                                                    Copy code

name, customer.last_name, category.name AS genre, COUNT(rental.rental_id) AS rental_count

r.customer_id = rental.customer_id
al.inventory_id = inventory.inventory_id
y.film_id = film.film_id
film.film_id = film_category.film_id
category.category_id = category.category_id
er_id = (
stomer_id

stomer.customer_id = rental.customer_id
ental.rental_id = payment.rental_id
customer_id
ent.amount) ASC

t_name, customer.last_name, genre;
```

Python Pandas:

```
python                                                                    Copy code

query = """
SELECT customer.first_name, customer.last_name, category.name AS genre, COUNT(rental.rental_id) AS rental_count
FROM customer
JOIN rental ON customer.customer_id = rental.customer_id
JOIN inventory ON rental.inventory_id = inventory.inventory_id
JOIN film ON inventory.film_id = film.film_id
JOIN film_category ON film.film_id = film_category.film_id
JOIN category ON film_category.category_id = category.category_id
WHERE customer.customer_id = (
    SELECT customer.customer_id
    FROM customer
    JOIN rental ON customer.customer_id = rental.customer_id
    JOIN payment ON rental.rental_id = payment.rental_id
    GROUP BY customer.customer_id
    ORDER BY SUM(payment.amount) ASC
    LIMIT 1
)
GROUP BY customer.first_name, customer.last_name, genre;
"""

df = pd.read_sql_query(query, conn)
print(df)
```

Continue generating

Soru 43: En çok kiralanan film hangi çalışan tarafından kiralanmış?

```
sql                                                                    Copy code

SELECT staff.first_name, staff.last_name, film.title, COUNT(rental.rental_id) AS rental_count
FROM staff
JOIN rental ON staff.staff_id = rental.staff_id
JOIN inventory ON rental.inventory_id = inventory.inventory_id
JOIN film ON inventory.film_id = film.film_id
WHERE film.film_id = (
    SELECT film.film_id
    FROM film
    JOIN inventory ON film.film_id = inventory.film_id
    JOIN rental ON inventory.inventory_id = rental.inventory_id
    GROUP BY film.film_id
    ORDER BY COUNT(rental.rental_id) DESC
    LIMIT 1
)
GROUP BY staff.first_name, staff.last_name, film.title;
```

Python Pandas:

```
python                                                                    Copy code

query = """
SELECT staff.first_name, staff.last_name, film.title, COUNT(rental.rental_id) AS rental_count
FROM staff
JOIN rental ON staff.staff_id = rental.staff_id
JOIN inventory ON rental.inventory_id = inventory.inventory_id
JOIN film ON inventory.film_id = film.film_id
WHERE film.film_id = (
    SELECT film.film_id
    FROM film
    JOIN inventory ON film.film_id = inventory.film_id
    JOIN rental ON inventory.inventory_id = rental.inventory_id
    GROUP BY film.film_id
    ORDER BY COUNT(rental.rental_id) DESC
    LIMIT 1
)
GROUP BY staff.first_name, staff.last_name, film.title;
"""

df = pd.read_sql_query(query, conn)
print(df)
```

Soru 44: En az kiralanan film hangi çalışan tarafından kiralanmış?

```
sql                                                                    Copy code

SELECT staff.first_name, staff.last_name, film.title, COUNT(rental.rental_id) AS rental_count
FROM staff
JOIN rental ON staff.staff_id = rental.staff_id
JOIN inventory ON rental.inventory_id = inventory.inventory_id
JOIN film ON inventory.film_id = film.film_id
WHERE film.film_id = (
    SELECT film.film_id
    FROM film
    JOIN inventory ON film.film_id = inventory.film_id
    JOIN rental ON inventory.inventory_id = rental.inventory_id
    GROUP BY film.film_id
    ORDER BY COUNT(rental.rental_id) ASC
    LIMIT 1
)
ORDER BY staff.first_name, staff.last_name, film.title;
```

Python Pandas:

```
python                                                                    Copy code

query = """
SELECT staff.first_name, staff.last_name, film.title, COUNT(rental.rental_id) AS rental_count
FROM staff
JOIN rental ON staff.staff_id = rental.staff_id
JOIN inventory ON rental.inventory_id = inventory.inventory_id
JOIN film ON inventory.film_id = film.film_id
WHERE film.film_id = (
    SELECT film.film_id
    FROM film
    JOIN inventory ON film.film_id = inventory.film_id
    JOIN rental ON inventory.inventory_id = rental.inventory_id
    GROUP BY film.film_id
    ORDER BY COUNT(rental.rental_id) ASC
    LIMIT 1
)
GROUP BY staff.first_name, staff.last_name, film.title;
"""

df = pd.read_sql_query(query, conn)
print(df)
```

Soru 45: En çok kazanç sağlayan film hangi çalışan tarafından kiralanmış?

```
sql                                                                    Copy code

SELECT staff.first_name, staff.last_name, film.title, SUM(payment.amount) AS total_revenue
FROM staff
JOIN rental ON staff.staff_id = rental.staff_id
JOIN inventory ON rental.inventory_id = inventory.inventory_id
JOIN film ON inventory.film_id = film.film_id
JOIN payment ON rental.rental_id = payment.rental_id
WHERE film.film_id = (
    SELECT film.film_id
    FROM film
    JOIN inventory ON film.film_id = inventory.film_id
    JOIN rental ON inventory.inventory_id = rental.inventory_id
    JOIN payment ON rental.rental_id = payment.rental_id
    GROUP BY film.film_id
    ORDER BY SUM(payment.amount) DESC
    LIMIT 1
)
GROUP BY staff.first_name, staff.last_name, film.title;
```

Python Pandas:

```
python                                                                    Copy code

query = """
SELECT staff.first_name, staff.last_name, film.title, SUM(payment.amount) AS total_revenue
FROM staff
JOIN rental ON staff.staff_id = rental.staff_id
JOIN inventory ON rental.inventory_id = inventory.inventory_id
JOIN film ON inventory.film_id = film.film_id
JOIN payment ON rental.rental_id = payment.rental_id
WHERE film.film_id = (
    SELECT film.film_id
    FROM film
    JOIN inventory ON film.film_id = inventory.film_id
    JOIN rental ON inventory.inventory_id = rental.inventory_id
    JOIN payment ON rental.rental_id = payment.rental_id
    GROUP BY film.film_id
    ORDER BY SUM(payment.amount) DESC
    LIMIT 1
)
GROUP BY staff.first_name, staff.last_name, film.title;
"""

df = pd.read_sql_query(query, conn)
print(df)
```

Soru 46: En az kazanç sağlayan film hangi çalışan tarafından kiralanmış?

sql

```
SELECT staff.first_name, staff.last_name, film.title, SUM(payment.amount) AS total_payment
FROM staff
JOIN rental ON staff.staff_id = rental.staff_id
JOIN inventory ON rental.inventory_id = inventory.inventory_id
JOIN film ON inventory.film_id = film.film_id
JOIN payment ON rental.rental_id = payment.rental_id
WHERE film.film_id = (
    SELECT film.film_id
    FROM film
    JOIN inventory ON film.film_id = inventory.film_id
    JOIN rental ON inventory.inventory_id = rental.inventory_id
    JOIN payment ON rental.rental_id = payment.rental_id
    GROUP BY film.film_id
    ORDER BY SUM(payment.amount) ASC
    LIMIT 1
)
GROUP BY staff.first_name, staff.last_name, film.title;
```

Soru 47: En çok kiralanan film hangi mağazada kiralanmış?

sql

```
SELECT store.store_id, film.title, COUNT(rental.rental_id) AS rental_count
FROM store
JOIN staff ON store.store_id = staff.store_id
JOIN rental ON staff.staff_id = rental.staff_id
JOIN inventory ON rental.inventory_id = inventory.inventory_id
JOIN film ON inventory.film_id = film.film_id
WHERE film.film_id = (
    SELECT film.film_id
    FROM film
    JOIN inventory ON film.film_id = inventory.film_id
    JOIN rental ON inventory.inventory_id = rental.inventory_id
    GROUP BY film.film_id
    ORDER BY COUNT(rental.rental_id) DESC
    LIMIT 1
)
GROUP BY store.store_id, film.title;
```

Soru 48: En az kiralanan film hangi mağazada kiralanmış?

sql

```
SELECT store.store_id, film.title, COUNT(rental.rental_id) AS rental_count
FROM store
JOIN staff ON store.store_id = staff.store_id
JOIN rental ON staff.staff_id = rental.staff_id
JOIN inventory ON rental.inventory_id = inventory.inventory_id
JOIN film ON inventory.film_id = film.film_id
WHERE film.film_id = (
    SELECT film.film_id
    FROM film
    JOIN inventory ON film.film_id = inventory.film_id
    JOIN rental ON inventory.inventory_id = rental.inventory_id
    GROUP BY film.film_id
    ORDER BY COUNT(rental.rental_id) ASC
    LIMIT 1
)
GROUP BY store.store_id, film.title;
```

Soru 49: En çok kazanç sağlayan film hangi mağazada kiralanmış?

sql

```
SELECT store.store_id, film.title, SUM(payment.amount) AS total_revenue
FROM store
JOIN staff ON store.store_id = staff.store_id
JOIN rental ON staff.staff_id = rental.staff_id
JOIN inventory ON rental.inventory_id = inventory.inventory_id
JOIN film ON inventory.film_id = film.film_id
JOIN payment ON rental.rental_id = payment.rental_id
WHERE film.film_id = (
    SELECT film.film_id
    FROM film
    JOIN inventory ON film.film_id = inventory.film_id
    JOIN rental ON inventory.inventory_id = rental.inventory_id
    JOIN payment ON rental.rental_id = payment.rental_id
    GROUP BY film.film_id
    ORDER BY SUM(payment.amount) DESC
    LIMIT 1
)
GROUP BY store.store_id, film.title;
```

Soru 50: En az kazanç sağlayan film hangi mağazada kiralanmış?

sql

```
SELECT store.store_id, film.title, SUM(payment.amount) AS total_revenue
FROM store
JOIN staff ON store.store_id = staff.store_id
JOIN rental ON staff.staff_id = rental.staff_id
JOIN inventory ON rental.inventory_id = inventory.inventory_id
JOIN film ON inventory.film_id = film.film_id
JOIN payment ON rental.rental_id = payment.rental_id
WHERE film.film_id = (
    SELECT film.film_id
    FROM film
    JOIN inventory ON film.film_id = inventory.film_id
    JOIN rental ON inventory.inventory_id = rental.inventory_id
    JOIN payment ON rental.rental_id = payment.rental_id
    GROUP BY film.film_id
    ORDER BY SUM(payment.amount) ASC
    LIMIT 1
)
GROUP BY store.store_id, film.title;
```

Soru 51: Müşterilerin kiraladıkları filmlerin toplam kiralama süresi ne kadar?

sql

```
me, SUM(julianday(rental.return_date) - julianday(rental.rental_date)) AS total_r  
.  
customer_id  
name;
```

Python Pandas:

python

```
query = """  
SELECT customer.first_name, customer.last_name, SUM(julianday(rental.return_date) - julianday(rental.rental_date)) AS total_rental_duration  
FROM customer  
JOIN rental ON customer.customer_id = rental.customer_id  
GROUP BY customer.first_name, customer.last_name;  
"""  
  
df = pd.read_sql_query(query, conn)  
print(df)
```

Soru 52: En çok kiralanan türdeki filmler hangileri?

sql

 Copy code

```
SELECT category.name AS genre, film.title, COUNT(rental.rental_id) AS rental_count
FROM film
JOIN film_category ON film.film_id = film_category.film_id
JOIN category ON film_category.category_id = category.category_id
JOIN inventory ON film.film_id = inventory.film_id
JOIN rental ON inventory.inventory_id = rental.inventory_id
GROUP BY genre, film.title
ORDER BY rental_count DESC
LIMIT 1;
```

Python Pandas:

python

 Copy code

```
query = """
SELECT category.name AS genre, film.title, COUNT(rental.rental_id) AS rental_count
FROM film
JOIN film_category ON film.film_id = film_category.film_id
JOIN category ON film_category.category_id = category.category_id
JOIN inventory ON film.film_id = inventory.film_id
JOIN rental ON inventory.inventory_id = rental.inventory_id
GROUP BY genre, film.title
ORDER BY rental_count DESC
LIMIT 1;
"""

df = pd.read_sql_query(query, conn)
print(df)
```



Soru 53: En az kiralanan türdeki filmler hangileri?

sql

Copy code

```
SELECT category.name AS genre, film.title, COUNT(rental.rental_id) AS rental_count
FROM film
JOIN film_category ON film.film_id = film_category.film_id
JOIN category ON film_category.category_id = category.category_id
JOIN inventory ON film.film_id = inventory.film_id
JOIN rental ON inventory.inventory_id = rental.inventory_id
GROUP BY genre, film.title
ORDER BY rental_count ASC
LIMIT 1;
```

Python Pandas:

python

Copy code

```
query = """
SELECT category.name AS genre, film.title, COUNT(rental.rental_id) AS rental_count
FROM film
JOIN film_category ON film.film_id = film_category.film_id
JOIN category ON film_category.category_id = category.category_id
JOIN inventory ON film.film_id = inventory.film_id
JOIN rental ON inventory.inventory_id = rental.inventory_id
GROUP BY genre, film.title
ORDER BY rental_count ASC
LIMIT 1;
"""

df = pd.read_sql_query(query, conn)
print(df)
```



Soru 54: Her müşteri için toplam ödeme miktarını bulun.

sql

 Copy code

```
SELECT customer.first_name, customer.last_name, SUM(payment.amount) AS total_payment
FROM customer
JOIN rental ON customer.customer_id = rental.customer_id
JOIN payment ON rental.rental_id = payment.rental_id
GROUP BY customer.first_name, customer.last_name;
```

Python Pandas:

python

 Copy code

```
query = """
SELECT customer.first_name, customer.last_name, SUM(payment.amount) AS total_payment
FROM customer
JOIN rental ON customer.customer_id = rental.customer_id
JOIN payment ON rental.rental_id = payment.rental_id
GROUP BY customer.first_name, customer.last_name;
"""

df = pd.read_sql_query(query, conn)
print(df)
```

Soru 55: Hangi filmler en uzun süre kiralanmış?

sql

Copy code

```
title, MAX(julianday(rental.return_date) - julianday(rental.rental_date)) AS max_rental_days
/ ON film.film_id = inventory.film_id
inventory.inventory_id = rental.inventory_id
.title
rental_days DESC;
```

Python Pandas:

python

Copy code

```
query = """
SELECT film.title, MAX(julianday(rental.return_date) - julianday(rental.rental_date)) AS
FROM film
JOIN inventory ON film.film_id = inventory.film_id
JOIN rental ON inventory.inventory_id = rental.inventory_id
GROUP BY film.title
ORDER BY max_rental_days DESC;
"""

df = pd.read_sql_query(query, conn)
print(df)
```



Soru 56: Hangi filmler en kısa süre kiralanmış?

sql

Copy code

```
title, MIN(julianday(rental.return_date) - julianday(rental.rental_date)) AS min_rental_days  
FROM film  
JOIN inventory ON film.film_id = inventory.film_id  
JOIN rental ON inventory.inventory_id = rental.inventory_id  
GROUP BY film.title  
ORDER BY min_rental_days ASC;
```

Python Pandas:

python

Copy code

```
query = """  
SELECT film.title, MIN(julianday(rental.return_date) - julianday(rental.rental_date)) AS min_rental_days  
FROM film  
JOIN inventory ON film.film_id = inventory.film_id  
JOIN rental ON inventory.inventory_id = rental.inventory_id  
GROUP BY film.title  
ORDER BY min_rental_days ASC;  
"""  
  
df = pd.read_sql_query(query, conn)  
print(df)
```

Soru 57: Müşterilerin kiraladığı filmler için ortalama ödeme miktarını bulun.

sql

Copy code

```
SELECT customer.first_name, customer.last_name, AVG(payment.amount) AS avg_payment
FROM customer
JOIN rental ON customer.customer_id = rental.customer_id
JOIN payment ON rental.rental_id = payment.rental_id
GROUP BY customer.first_name, customer.last_name;
```

Python Pandas:

python

Copy code

```
query = """
SELECT customer.first_name, customer.last_name, AVG(payment.amount) AS avg_payment
FROM customer
JOIN rental ON customer.customer_id = rental.customer_id
JOIN payment ON rental.rental_id = payment.rental_id
GROUP BY customer.first_name, customer.last_name;
"""

df = pd.read_sql_query(query, conn)
print(df)
```



Soru 58: En çok kiralanan filmlerin ortalama kiralama süresi nedir?

sql

Copy code

```
title, AVG(julianday(rental.return_date) - julianday(rental.rental_date)) AS avg_rental_days
FROM film
JOIN inventory ON film.film_id = inventory.film_id
JOIN rental ON inventory.inventory_id = rental.inventory_id
GROUP BY title
ORDER BY COUNT(rental.rental_id) DESC
```

Python Pandas:

python

Copy code

```
query = """
SELECT film.title, AVG(julianday(rental.return_date) - julianday(rental.rental_date)) AS avg_rental_days
FROM film
JOIN inventory ON film.film_id = inventory.film_id
JOIN rental ON inventory.inventory_id = rental.inventory_id
GROUP BY film.title
ORDER BY COUNT(rental.rental_id) DESC
LIMIT 10;
"""

df = pd.read_sql_query(query, conn)
print(df)
```



Soru 59: En az kiralanan filmlerin ortalama kiralama süresi nedir?

```
title, AVG(julianday(rental.return_date) - julianday(rental.rental_date)) AS avg_rental_days  
/ ON film.film_id = inventory.film_id  
| inventory.inventory_id = rental.inventory_id  
.title  
[(rental.rental_id) ASC
```

Python Pandas:

python

Copy code

```
query = """  
SELECT film.title, AVG(julianday(rental.return_date) - julianday(rental.rental_date)) AS  
FROM film  
JOIN inventory ON film.film_id = inventory.film_id  
JOIN rental ON inventory.inventory_id = rental.inventory_id  
GROUP BY film.title  
ORDER BY COUNT(rental.rental_id) ASC  
LIMIT 10;  
"""  
  
df = pd.read_sql_query(query, conn)  
print(df)
```

Soru 60: En çok kazanç sağlayan müşterilerin ortalama ödeme miktarı nedir?

sql

 Copy code

```
SELECT customer.first_name, customer.last_name, AVG(payment.amount) AS avg_payment
FROM customer
JOIN rental ON customer.customer_id = rental.customer_id
JOIN payment ON rental.rental_id = payment.rental_id
GROUP BY customer.first_name, customer.last_name
ORDER BY SUM(payment.amount) DESC
LIMIT 10;
```

Python Pandas:

python

 Copy code

```
query = """
SELECT customer.first_name, customer.last_name, AVG(payment.amount) AS avg_payment
FROM customer
JOIN rental ON customer.customer_id = rental.customer_id
JOIN payment ON rental.rental_id = payment.rental_id
GROUP BY customer.first_name, customer.last_name
ORDER BY SUM(payment.amount) DESC
LIMIT 10;
"""

df = pd.read_sql_query(query, conn)
print(df)
```



Soru 61: En az kazanç sağlayan müşterilerin ortalama ödeme miktarı nedir?

sql

Copy code

```
SELECT customer.first_name, customer.last_name, AVG(payment.amount) AS avg_payment
FROM customer
JOIN rental ON customer.customer_id = rental.customer_id
JOIN payment ON rental.rental_id = payment.rental_id
GROUP BY customer.first_name, customer.last_name
ORDER BY SUM(payment.amount) ASC
LIMIT 10;
```

Python Pandas:

python

Copy code

```
query = """
SELECT customer.first_name, customer.last_name, AVG(payment.amount) AS avg_payment
FROM customer
JOIN rental ON customer.customer_id = rental.customer_id
JOIN payment ON rental.rental_id = payment.rental_id
GROUP BY customer.first_name, customer.last_name
ORDER BY SUM(payment.amount) ASC
LIMIT 10;
"""

df = pd.read_sql_query(query, conn)
print(df)
```

Soru 62: Mağazalardaki toplam kiralama süresini bulun.

sql

Copy code

```
store_id, SUM(julianday(rental.return_date) - julianday(rental.rental_date)) AS total_rental_days
FROM store
JOIN staff ON store.store_id = staff.store_id
JOIN rental ON staff.staff_id = rental.staff_id
GROUP BY store_id;
```

Python Pandas:

python

Copy code

```
query = """
SELECT store.store_id, SUM(julianday(rental.return_date) - julianday(rental.rental_date))
FROM store
JOIN staff ON store.store_id = staff.store_id
JOIN rental ON staff.staff_id = rental.staff_id
GROUP BY store.store_id;
"""

df = pd.read_sql_query(query, conn)
print(df)
```

Soru 63: En uzun süre kiralanan film hangi mağazada kiralanmış?

sql

Copy code

```
title, MAX(julianday(rental.return_date) - julianday(rental.rental_date)) AS max_rental_days  
  
    staff.store_id  
= rental.staff_id  
tory_id = inventory.inventory_id  
    = film.film_id  
title
```

Python Pandas:

python

Copy code

```
query = """  
SELECT store.store_id, film.title, MAX(julianday(rental.return_date) - julianday(rental.rental_date)) AS max_rental_days  
FROM store  
JOIN staff ON store.store_id = staff.store_id  
JOIN rental ON staff.staff_id = rental.staff_id  
JOIN inventory ON rental.inventory_id = inventory.inventory_id  
JOIN film ON inventory.film_id = film.film_id  
GROUP BY store.store_id, film.title  
ORDER BY max_rental_days DESC  
LIMIT 1;  
"""  
  
df = pd.read_sql_query(query, conn)  
print(df)
```

Soru 64: En kısa süre kiralanan film hangi mağazada kiralanmış?

sql

Copy code

```
tle, MIN(julianday(rental.return_date) - julianday(rental.rental_date)) AS min_rental_days

    staff.store_id
= rental.staff_id
tory_id = inventory.inventory_id
    = film.film_id
title
```

Python Pandas:

python

Copy code

```
query = """
SELECT store.store_id, film.title, MIN(julianday(rental.return_date) - julianday(rental.
FROM store
JOIN staff ON store.store_id = staff.store_id
JOIN rental ON staff.staff_id = rental.staff_id
JOIN inventory ON rental.inventory_id = inventory.inventory_id
JOIN film ON inventory.film_id = film.film_id
GROUP BY store.store_id, film.title
ORDER BY min_rental_days ASC
LIMIT 1;
"""

df = pd.read_sql_query(query, conn)
print(df)
```

Soru 65: Her filmin ortalama kiralama süresi nedir?

sql

Copy code

```
title, AVG(julianday(rental.return_date) - julianday(rental.rental_date)) AS avg_rental_days  
  
ON film.film_id = inventory.film_id  
inventory.inventory_id = rental.inventory_id  
title;
```

Python Pandas:

python

Copy code

```
query = """  
SELECT film.title, AVG(julianday(rental.return_date) - julianday(rental.rental_date)) AS  
FROM film  
JOIN inventory ON film.film_id = inventory.film_id  
JOIN rental ON inventory.inventory_id = rental.inventory_id  
GROUP BY film.title;  
"""  
  
df = pd.read_sql_query(query, conn)  
print(df)
```

Soru 66: Hangi filmler en çok kiralanan kategoride?

sql

 Copy code

```
SELECT category.name AS genre, film.title, COUNT(rental.rental_id) AS rental_count
FROM category
JOIN film_category ON category.category_id = film_category.category_id
JOIN film ON film_category.film_id = film.film_id
JOIN inventory ON film.film_id = inventory.film_id
JOIN rental ON inventory.inventory_id = rental.inventory_id
GROUP BY genre, film.title
ORDER BY rental_count DESC;
```

Python Pandas:

python

 Copy code

```
query = """
SELECT category.name AS genre, film.title, COUNT(rental.rental_id) AS rental_count
FROM category
JOIN film_category ON category.category_id = film_category.category_id
JOIN film ON film_category.film_id = film.film_id
JOIN inventory ON film.film_id = inventory.film_id
JOIN rental ON inventory.inventory_id = rental.inventory_id
GROUP BY genre, film.title
ORDER BY rental_count DESC;
"""

df = pd.read_sql_query(query, conn)
print(df)
```



Soru 67: Hangi filmler en az kiralanan kategoride?

sql

 Copy code

```
SELECT category.name AS genre, film.title, COUNT(rental.rental_id) AS rental_count
FROM category
JOIN film_category ON category.category_id = film_category.category_id
JOIN film ON film_category.film_id = film.film_id
JOIN inventory ON film.film_id = inventory.film_id
JOIN rental ON inventory.inventory_id = rental.inventory_id
GROUP BY genre, film.title
ORDER BY rental_count ASC;
```

Python Pandas:

python

 Copy code

```
query = """
SELECT category.name AS genre, film.title, COUNT(rental.rental_id) AS rental_count
FROM category
JOIN film_category ON category.category_id = film_category.category_id
JOIN film ON film_category.film_id = film.film_id
JOIN inventory ON film.film_id = inventory.film_id
JOIN rental ON inventory.inventory_id = rental.inventory_id
GROUP BY genre, film.title
ORDER BY rental_count ASC;
"""

df = pd.read_sql_query(query, conn)
print(df)
```



Soru 68: Hangi mağazalarda en çok film kiralanmış?

sql

 Copy code

```
SELECT store.store_id, COUNT(rental.rental_id) AS rental_count
FROM store
JOIN staff ON store.store_id = staff.store_id
JOIN rental ON staff.staff_id = rental.staff_id
GROUP BY store.store_id
ORDER BY rental_count DESC;
```

Python Pandas:

python

 Copy code

```
query = """
SELECT store.store_id, COUNT(rental.rental_id) AS rental_count
FROM store
JOIN staff ON store.store_id = staff.store_id
JOIN rental ON staff.staff_id = rental.staff_id
GROUP BY store.store_id
ORDER BY rental_count DESC;
"""

df = pd.read_sql_query(query, conn)
print(df)
```

Soru 69: Hangi mağazalarda en az film kiralanmış?

sql

 Copy code

```
SELECT store.store_id, COUNT(rental.rental_id) AS rental_count
FROM store
JOIN staff ON store.store_id = staff.store_id
JOIN rental ON staff.staff_id = rental.staff_id
GROUP BY store.store_id
ORDER BY rental_count ASC;
```

Python Pandas:

python

 Copy code

```
query = """
SELECT store.store_id, COUNT(rental.rental_id) AS rental_count
FROM store
JOIN staff ON store.store_id = staff.store_id
JOIN rental ON staff.staff_id = rental.staff_id
GROUP BY store.store_id
ORDER BY rental_count ASC;
"""

df = pd.read_sql_query(query, conn)
print(df)
```

Soru 70: Hangi aktörler en çok filmde rol almış?

sql

 Copy code

```
SELECT actor.first_name, actor.last_name, COUNT(film_actor.film_id) AS film_count
FROM actor
JOIN film_actor ON actor.actor_id = film_actor.actor_id
GROUP BY actor.first_name, actor.last_name
ORDER BY film_count DESC;
```

Python Pandas:

python

 Copy code

```
query = """
SELECT actor.first_name, actor.last_name, COUNT(film_actor.film_id) AS film_count
FROM actor
JOIN film_actor ON actor.actor_id = film_actor.actor_id
GROUP BY actor.first_name, actor.last_name
ORDER BY film_count DESC;
"""

df = pd.read_sql_query(query, conn)
print(df)
```

Soru 71: Hangi aktörler en az filmde rol almış?

sql

 Copy code

```
SELECT actor.first_name, actor.last_name, COUNT(film_actor.film_id) AS film_count
FROM actor
JOIN film_actor ON actor.actor_id = film_actor.actor_id
GROUP BY actor.first_name, actor.last_name
ORDER BY film_count ASC;
```

Python Pandas:

python

 Copy code

```
query = """
SELECT actor.first_name, actor.last_name, COUNT(film_actor.film_id) AS film_count
FROM actor
JOIN film_actor ON actor.actor_id = film_actor.actor_id
GROUP BY actor.first_name, actor.last_name
ORDER BY film_count ASC;
"""

df = pd.read_sql_query(query, conn)
print(df)
```

Soru 72: Hangi aktörlerin oynadığı filmler en çok kiralanmış?

sql

 Copy code

```
SELECT actor.first_name, actor.last_name, COUNT(rental.rental_id) AS rental_count
FROM actor
JOIN film_actor ON actor.actor_id = film_actor.actor_id
JOIN film ON film_actor.film_id = film.film_id
JOIN inventory ON film.film_id = inventory.film_id
JOIN rental ON inventory.inventory_id = rental.inventory_id
GROUP BY actor.first_name, actor.last_name
ORDER BY rental_count DESC;
```

Python Pandas:

python

 Copy code

```
query = """
SELECT actor.first_name, actor.last_name, COUNT(rental.rental_id) AS rental_count
FROM actor
JOIN film_actor ON actor.actor_id = film_actor.actor_id
JOIN film ON film_actor.film_id = film.film_id
JOIN inventory ON film.film_id = inventory.film_id
JOIN rental ON inventory.inventory_id = rental.inventory_id
GROUP BY actor.first_name, actor.last_name
ORDER BY rental_count DESC;
"""

df = pd.read_sql_query(query, conn)
print(df)
```

Soru 73: Hangi aktörlerin oynadığı filmler en az kiralanmış?

sql

Copy code

```
SELECT actor.first_name, actor.last_name, COUNT(rental.rental_id) AS rental_count
FROM actor
JOIN film_actor ON actor.actor_id = film_actor.actor_id
JOIN film ON film_actor.film_id = film.film_id
JOIN inventory ON film.film_id = inventory.film_id
JOIN rental ON inventory.inventory_id = rental.inventory_id
GROUP BY actor.first_name, actor.last_name
ORDER BY rental_count ASC;
```

Python Pandas:

python

Copy code

```
query = """
SELECT actor.first_name, actor.last_name, COUNT(rental.rental_id) AS rental_count
FROM actor
JOIN film_actor ON actor.actor_id = film_actor.actor_id
JOIN film ON film_actor.film_id = film.film_id
JOIN inventory ON film.film_id = inventory.film_id
JOIN rental ON inventory.inventory_id = rental.inventory_id
GROUP BY actor.first_name, actor.last_name
ORDER BY rental_count ASC;
"""

df = pd.read_sql_query(query, conn)
print(df)
```



Soru 74: Hangi kategorilerde en fazla aktör oynamış?

sql

 Copy code

```
SELECT category.name AS genre, COUNT(DISTINCT film_actor.actor_id) AS actor_count
FROM category
JOIN film_category ON category.category_id = film_category.category_id
JOIN film ON film_category.film_id = film.film_id
JOIN film_actor ON film.film_id = film_actor.film_id
GROUP BY genre
ORDER BY actor_count DESC;
```

Python Pandas:

python

 Copy code

```
query = """
SELECT category.name AS genre, COUNT(DISTINCT film_actor.actor_id) AS actor_count
FROM category
JOIN film_category ON category.category_id = film_category.category_id
JOIN film ON film_category.film_id = film.film_id
JOIN film_actor ON film.film_id = film_actor.film_id
GROUP BY genre
ORDER BY actor_count DESC;
"""

df = pd.read_sql_query(query, conn)
print(df)
```



Soru 75: Hangi kategorilerde en az aktör oynamış?

sql

 Copy code

```
SELECT category.name AS genre, COUNT(DISTINCT film_actor.actor_id) AS actor_count
FROM category
JOIN film_category ON category.category_id = film_category.category_id
JOIN film ON film_category.film_id = film.film_id
JOIN film_actor ON film.film_id = film_actor.film_id
GROUP BY genre
ORDER BY actor_count ASC;
```

Python Pandas:

python

 Copy code

```
query = """
SELECT category.name AS genre, COUNT(DISTINCT film_actor.actor_id) AS actor_count
FROM category
JOIN film_category ON category.category_id = film_category.category_id
JOIN film ON film_category.film_id = film.film_id
JOIN film_actor ON film.film_id = film_actor.film_id
GROUP BY genre
ORDER BY actor_count ASC;
"""

df = pd.read_sql_query(query, conn)
print(df)
```



Soru 76: Hangi kategorilerde oynayan filmler en çok kiralanmış?

SQL:

sql

Copy code

```
SELECT category.name AS genre, COUNT(rental.rental_id) AS rental_count
FROM category
JOIN film_category ON category.category_id = film_category.category_id
JOIN film ON film_category.film_id = film.film_id
JOIN inventory ON film.film_id = inventory.film_id
JOIN rental ON inventory.inventory_id = rental.inventory_id
GROUP BY genre
ORDER BY rental_count DESC;
```

Python Pandas:

python

Copy code

```
query = """
SELECT category.name AS genre, COUNT(rental.rental_id) AS rental_count
FROM category
JOIN film_category ON category.category_id = film_category.category_id
JOIN film ON film_category.film_id = film.film_id
JOIN inventory ON film.film_id = inventory.film_id
JOIN rental ON inventory.inventory_id = rental.inventory_id
GROUP BY genre
ORDER BY rental_count DESC;
"""

df = pd.read_sql_query(query, conn)
print(df)
```



Soru 77: Hangi kategorilerde oynayan filmler en az kiralanmış?

sql

 Copy code

```
SELECT category.name AS genre, COUNT(rental.rental_id) AS rental_count
FROM category
JOIN film_category ON category.category_id = film_category.category_id
JOIN film ON film_category.film_id = film.film_id
JOIN inventory ON film.film_id = inventory.film_id
JOIN rental ON inventory.inventory_id = rental.inventory_id
GROUP BY genre
ORDER BY rental_count ASC;
```

Python Pandas:

python

 Copy code

```
query = """
SELECT category.name AS genre, COUNT(rental.rental_id) AS rental_count
FROM category
JOIN film_category ON category.category_id = film_category.category_id
JOIN film ON film_category.film_id = film.film_id
JOIN inventory ON film.film_id = inventory.film_id
JOIN rental ON inventory.inventory_id = rental.inventory_id
GROUP BY genre
ORDER BY rental_count ASC;
"""

df = pd.read_sql_query(query, conn)
print(df)
```

Soru 78: En fazla kazanç sağlayan kategoriler hangileri?

```
SELECT category.name AS genre, SUM(payment.amount) AS total_revenue
FROM category
JOIN film_category ON category.category_id = film_category.category_id
JOIN film ON film_category.film_id = film.film_id
JOIN inventory ON film.film_id = inventory.film_id
JOIN rental ON inventory.inventory_id = rental.inventory_id
JOIN payment ON rental.rental_id = payment.rental_id
GROUP BY genre
ORDER BY total_revenue DESC;
```

Python Pandas:

python

 Copy code

```
query = """
SELECT category.name AS genre, SUM(payment.amount) AS total_revenue
FROM category
JOIN film_category ON category.category_id = film_category.category_id
JOIN film ON film_category.film_id = film.film_id
JOIN inventory ON film.film_id = inventory.film_id
JOIN rental ON inventory.inventory_id = rental.inventory_id
JOIN payment ON rental.rental_id = payment.rental_id
GROUP BY genre
ORDER BY total_revenue DESC;
"""

df = pd.read_sql_query(query, conn)
print(df)
```



Soru 79: En az kazanç sağlayan kategoriler hangileri?

sql

 Copy code

```
SELECT category.name AS genre, SUM(payment.amount) AS total_revenue
FROM category
JOIN film_category ON category.category_id = film_category.category_id
JOIN film ON film_category.film_id = film.film_id
JOIN inventory ON film.film_id = inventory.film_id
JOIN rental ON inventory.inventory_id = rental.inventory_id
JOIN payment ON rental.rental_id = payment.rental_id
GROUP BY genre
ORDER BY total_revenue ASC;
```

Python Pandas:

python

 Copy code

```
query = """
SELECT category.name AS genre, SUM(payment.amount) AS total_revenue
FROM category
JOIN film_category ON category.category_id = film_category.category_id
JOIN film ON film_category.film_id = film.film_id
JOIN inventory ON film.film_id = inventory.film_id
JOIN rental ON inventory.inventory_id = rental.inventory_id
JOIN payment ON rental.rental_id = payment.rental_id
GROUP BY genre
ORDER BY total_revenue ASC;
"""

df = pd.read_sql_query(query, conn)
print(df)
```



Soru 80: En çok film kiralayan müşterilerin ortalama ödeme miktarı nedir?

sql

 Copy code

```
SELECT customer.first_name, customer.last_name, AVG(payment.amount) AS avg_payment
FROM customer
JOIN rental ON customer.customer_id = rental.customer_id
JOIN payment ON rental.rental_id = payment.rental_id
GROUP BY customer.first_name, customer.last_name
ORDER BY COUNT(rental.rental_id) DESC
LIMIT 10;
```

Python Pandas:

python

 Copy code

```
query = """
SELECT customer.first_name, customer.last_name, AVG(payment.amount) AS avg_payment
FROM customer
JOIN rental ON customer.customer_id = rental.customer_id
JOIN payment ON rental.rental_id = payment.rental_id
GROUP BY customer.first_name, customer.last_name
ORDER BY COUNT(rental.rental_id) DESC
LIMIT 10;
"""

df = pd.read_sql_query(query, conn)
print(df)
```



Soru 81: En az film kiralayan müşterilerin ortalama ödeme miktarı nedir?

sql

 Copy code

```
SELECT customer.first_name, customer.last_name, AVG(payment.amount) AS avg_payment
FROM customer
JOIN rental ON customer.customer_id = rental.customer_id
JOIN payment ON rental.rental_id = payment.rental_id
GROUP BY customer.first_name, customer.last_name
ORDER BY COUNT(rental.rental_id) ASC
LIMIT 10;
```

Python Pandas:

python

 Copy code

```
query = """
SELECT customer.first_name, customer.last_name, AVG(payment.amount) AS avg_payment
FROM customer
JOIN rental ON customer.customer_id = rental.customer_id
JOIN payment ON rental.rental_id = payment.rental_id
GROUP BY customer.first_name, customer.last_name
ORDER BY COUNT(rental.rental_id) ASC
LIMIT 10;
"""

df = pd.read_sql_query(query, conn)
print(df)
```

Soru 82: Hangi mağazalarda hangi kategoriler en çok kiralanmış?

```
SELECT store.store_id, category.name AS genre, COUNT(rental.rental_id) AS rental_count
FROM store
JOIN staff ON store.store_id = staff.store_id
JOIN rental ON staff.staff_id = rental.staff_id
JOIN inventory ON rental.inventory_id = inventory.inventory_id
JOIN film ON inventory.film_id = film.film_id
JOIN film_category ON film.film_id = film_category.film_id
JOIN category ON film_category.category_id = category.category_id
GROUP BY store.store_id, genre
ORDER BY rental_count DESC;
```

Python Pandas:

python

 Copy code

```
query = """
SELECT store.store_id, category.name AS genre, COUNT(rental.rental_id) AS rental_count
FROM store
JOIN staff ON store.store_id = staff.store_id
JOIN rental ON staff.staff_id = rental.staff_id
JOIN inventory ON rental.inventory_id = inventory.inventory_id
JOIN film ON inventory.film_id = film.film_id
JOIN film_category ON film.film_id = film_category.film_id
JOIN category ON film_category.category_id = category.category_id
GROUP BY store.store_id, genre
ORDER BY rental_count DESC;
"""

df = pd.read_sql_query(query, conn)
print(df)
```



Soru 83: Hangi mağazalarda hangi kategoriler en az kiralanmış?

```
SELECT store.store_id, category.name AS genre, COUNT(rental.rental_id) AS rental_count
FROM store
JOIN staff ON store.store_id = staff.store_id
JOIN rental ON staff.staff_id = rental.staff_id
JOIN inventory ON rental.inventory_id = inventory.inventory_id
JOIN film ON inventory.film_id = film.film_id
JOIN film_category ON film.film_id = film_category.film_id
JOIN category ON film_category.category_id = category.category_id
GROUP BY store.store_id, genre
ORDER BY rental_count ASC;
```

Python Pandas:

python

 Copy code

```
query = """
SELECT store.store_id, category.name AS genre, COUNT(rental.rental_id) AS rental_count
FROM store
JOIN staff ON store.store_id = staff.store_id
JOIN rental ON staff.staff_id = rental.staff_id
JOIN inventory ON rental.inventory_id = inventory.inventory_id
JOIN film ON inventory.film_id = film.film_id
JOIN film_category ON film.film_id = film_category.film_id
JOIN category ON film_category.category_id = category.category_id
GROUP BY store.store_id, genre
ORDER BY rental_count ASC;
"""

df = pd.read_sql_query(query, conn)
print(df)
```

Soru 84: En fazla sayıda film kiralayan mağaza hangisidir ve toplam kiralama sayısı nedir?

sql

 Copy code

```
SELECT store.store_id, COUNT(rental.rental_id) AS total_rentals
FROM store
JOIN staff ON store.store_id = staff.store_id
JOIN rental ON staff.staff_id = rental.staff_id
GROUP BY store.store_id
ORDER BY total_rentals DESC
LIMIT 1;
```

Python Pandas:

python

 Copy code

```
query = """
SELECT store.store_id, COUNT(rental.rental_id) AS total_rentals
FROM store
JOIN staff ON store.store_id = staff.store_id
JOIN rental ON staff.staff_id = rental.staff_id
GROUP BY store.store_id
ORDER BY total_rentals DESC
LIMIT 1;
"""

df = pd.read_sql_query(query, conn)
print(df)
```

Soru 85: En az sayıda film kiralayan mağaza hangisidir ve toplam kiralama sayısı nedir?

sql

 Copy code

```
SELECT store.store_id, COUNT(rental.rental_id) AS total_rentals
FROM store
JOIN staff ON store.store_id = staff.store_id
JOIN rental ON staff.staff_id = rental.staff_id
GROUP BY store.store_id
ORDER BY total_rentals ASC
LIMIT 1;
```

Python Pandas:

python

 Copy code

```
query = """
SELECT store.store_id, COUNT(rental.rental_id) AS total_rentals
FROM store
JOIN staff ON store.store_id = staff.store_id
JOIN rental ON staff.staff_id = rental.staff_id
GROUP BY store.store_id
ORDER BY total_rentals ASC
LIMIT 1;
"""

df = pd.read_sql_query(query, conn)
print(df)
```

Soru 86: En fazla sayıda kiralama yapan müşteri hangisidir ve toplam kiralama sayısı nedir?

sql

Copy code

```
SELECT customer.first_name, customer.last_name, COUNT(rental.rental_id) AS total_rentals
FROM customer
JOIN rental ON customer.customer_id = rental.customer_id
GROUP BY customer.first_name, customer.last_name
ORDER BY total_rentals DESC
LIMIT 1;
```

Python Pandas:

python

Copy code

```
query = """
SELECT customer.first_name, customer.last_name, COUNT(rental.rental_id) AS total_rentals
FROM customer
JOIN rental ON customer.customer_id = rental.customer_id
GROUP BY customer.first_name, customer.last_name
ORDER BY total_rentals DESC
LIMIT 1;
"""

df = pd.read_sql_query(query, conn)
print(df)
```

Soru 87: En az sayıda kiralama yapan müşteri hangisidir ve toplam kiralama sayısı nedir?

sql

 Copy code

```
SELECT customer.first_name, customer.last_name, COUNT(rental.rental_id) AS total_rentals
FROM customer
JOIN rental ON customer.customer_id = rental.customer_id
GROUP BY customer.first_name, customer.last_name
ORDER BY total_rentals ASC
LIMIT 1;
```

Python Pandas:

python

 Copy code

```
query = """
SELECT customer.first_name, customer.last_name, COUNT(rental.rental_id) AS total_rentals
FROM customer
JOIN rental ON customer.customer_id = rental.customer_id
GROUP BY customer.first_name, customer.last_name
ORDER BY total_rentals ASC
LIMIT 1;
"""

df = pd.read_sql_query(query, conn)
print(df)
```

Soru 88: En fazla sayıda kiralanan film hangisidir ve toplam kiralama sayısı nedir?

sql

 Copy code

```
SELECT film.title, COUNT(rental.rental_id) AS total_rentals
FROM film
JOIN inventory ON film.film_id = inventory.film_id
JOIN rental ON inventory.inventory_id = rental.inventory_id
GROUP BY film.title
ORDER BY total_rentals DESC
LIMIT 1;
```

Python Pandas:

python

 Copy code

```
query = """
SELECT film.title, COUNT(rental.rental_id) AS total_rentals
FROM film
JOIN inventory ON film.film_id = inventory.film_id
JOIN rental ON inventory.inventory_id = rental.inventory_id
GROUP BY film.title
ORDER BY total_rentals DESC
LIMIT 1;
"""

df = pd.read_sql_query(query, conn)
print(df)
```

Soru 89: En az sayıda kiralanan film hangisidir ve toplam kiralama sayısı nedir?

sql

 Copy code

```
SELECT film.title, COUNT(rental.rental_id) AS total_rentals
FROM film
JOIN inventory ON film.film_id = inventory.film_id
JOIN rental ON inventory.inventory_id = rental.inventory_id
GROUP BY film.title
ORDER BY total_rentals ASC
LIMIT 1;
```

Python Pandas:

python

 Copy code

```
query = """
SELECT film.title, COUNT(rental.rental_id) AS total_rentals
FROM film
JOIN inventory ON film.film_id = inventory.film_id
JOIN rental ON inventory.inventory_id = rental.inventory_id
GROUP BY film.title
ORDER BY total_rentals ASC
LIMIT 1;
"""

df = pd.read_sql_query(query, conn)
print(df)
```

Soru 90: En fazla kazanç sağlayan müşteri hangisidir ve toplam kazancı nedir?

sql

 Copy code

```
SELECT customer.first_name, customer.last_name, SUM(payment.amount) AS total_revenue
FROM customer
JOIN rental ON customer.customer_id = rental.customer_id
JOIN payment ON rental.rental_id = payment.rental_id
GROUP BY customer.first_name, customer.last_name
ORDER BY total_revenue DESC
LIMIT 1;
```

Python Pandas:

python

 Copy code

```
query = """
SELECT customer.first_name, customer.last_name, SUM(payment.amount) AS total_revenue
FROM customer
JOIN rental ON customer.customer_id = rental.customer_id
JOIN payment ON rental.rental_id = payment.rental_id
GROUP BY customer.first_name, customer.last_name
ORDER BY total_revenue DESC
LIMIT 1;
"""

df = pd.read_sql_query(query, conn)
print(df)
```

Soru 91: En az kazanç sağlayan müşteri hangisidir ve toplam kazancı nedir?

sql

 Copy code

```
SELECT customer.first_name, customer.last_name, SUM(payment.amount) AS total_revenue
FROM customer
JOIN rental ON customer.customer_id = rental.customer_id
JOIN payment ON rental.rental_id = payment.rental_id
GROUP BY customer.first_name, customer.last_name
ORDER BY total_revenue ASC
LIMIT 1;
```

Python Pandas:

python

 Copy code

```
query = """
SELECT customer.first_name, customer.last_name, SUM(payment.amount) AS total_revenue
FROM customer
JOIN rental ON customer.customer_id = rental.customer_id
JOIN payment ON rental.rental_id = payment.rental_id
GROUP BY customer.first_name, customer.last_name
ORDER BY total_revenue ASC
LIMIT 1;
"""

df = pd.read_sql_query(query, conn)
print(df)
```

Soru 92: Hangi film, hangi kategoride en çok kazanç sağlamış?

```
SELECT film.title, category.name AS genre, SUM(payment.amount) AS total_revenue
FROM film
JOIN film_category ON film.film_id = film_category.film_id
JOIN category ON film_category.category_id = category.category_id
JOIN inventory ON film.film_id = inventory.film_id
JOIN rental ON inventory.inventory_id = rental.inventory_id
JOIN payment ON rental.rental_id = payment.rental_id
GROUP BY film.title, genre
ORDER BY total_revenue DESC
LIMIT 1;
```

Python Pandas:

python

 Copy code

```
query = """
SELECT film.title, category.name AS genre, SUM(payment.amount) AS total_revenue
FROM film
JOIN film_category ON film.film_id = film_category.film_id
JOIN category ON film_category.category_id = category.category_id
JOIN inventory ON film.film_id = inventory.film_id
JOIN rental ON inventory.inventory_id = rental.inventory_id
JOIN payment ON rental.rental_id = payment.rental_id
GROUP BY film.title, genre
ORDER BY total_revenue DESC
LIMIT 1;
"""

df = pd.read_sql_query(query, conn)
print(df)
```



Soru 93: Hangi film, hangi kategoride en az kazanç sağlamış?

```
SELECT film.title, category.name AS genre, SUM(payment.amount) AS total_revenue
FROM film
JOIN film_category ON film.film_id = film_category.film_id
JOIN category ON film_category.category_id = category.category_id
JOIN inventory ON film.film_id = inventory.film_id
JOIN rental ON inventory.inventory_id = rental.inventory_id
JOIN payment ON rental.rental_id = payment.rental_id
GROUP BY film.title, genre
ORDER BY total_revenue ASC
LIMIT 1;
```

Python Pandas:

python

 Copy code

```
query = """
SELECT film.title, category.name AS genre, SUM(payment.amount) AS total_revenue
FROM film
JOIN film_category ON film.film_id = film_category.film_id
JOIN category ON film_category.category_id = category.category_id
JOIN inventory ON film.film_id = inventory.film_id
JOIN rental ON inventory.inventory_id = rental.inventory_id
JOIN payment ON rental.rental_id = payment.rental_id
GROUP BY film.title, genre
ORDER BY total_revenue ASC
LIMIT 1;
"""

df = pd.read_sql_query(query, conn)
print(df)
```



Soru 94: En fazla kazanç sağlayan mağaza hangisidir ve toplam kazancı nedir?

sql

 Copy code

```
SELECT store.store_id, SUM(payment.amount) AS total_revenue
FROM store
JOIN staff ON store.store_id = staff.store_id
JOIN rental ON staff.staff_id = rental.staff_id
JOIN payment ON rental.rental_id = payment.rental_id
GROUP BY store.store_id
ORDER BY total_revenue DESC
LIMIT 1;
```

Python Pandas:

python

 Copy code

```
query = """
SELECT store.store_id, SUM(payment.amount) AS total_revenue
FROM store
JOIN staff ON store.store_id = staff.store_id
JOIN rental ON staff.staff_id = rental.staff_id
JOIN payment ON rental.rental_id = payment.rental_id
GROUP BY store.store_id
ORDER BY total_revenue DESC
LIMIT 1;
"""

df = pd.read_sql_query(query, conn)
print(df)
```



Soru 95: En az kazanç sağlayan mağaza hangisidir ve toplam kazancı nedir?

sql

 Copy code

```
SELECT store.store_id, SUM(payment.amount) AS total_revenue
FROM store
JOIN staff ON store.store_id = staff.store_id
JOIN rental ON staff.staff_id = rental.staff_id
JOIN payment ON rental.rental_id = payment.rental_id
GROUP BY store.store_id
ORDER BY total_revenue ASC
LIMIT 1;
```

Python Pandas:

python

 Copy code

```
query = """
SELECT store.store_id, SUM(payment.amount) AS total_revenue
FROM store
JOIN staff ON store.store_id = staff.store_id
JOIN rental ON staff.staff_id = rental.staff_id
JOIN payment ON rental.rental_id = payment.rental_id
GROUP BY store.store_id
ORDER BY total_revenue ASC
LIMIT 1;
"""

df = pd.read_sql_query(query, conn)
print(df)
```

Soru 96: En fazla sayıda farklı film kiralayan müşteri hangisidir?

sql

Copy code

```
customer.first_name, customer.last_name, COUNT(DISTINCT rental.inventory_id) AS unique_rentals
FROM customer
JOIN rental ON customer.customer_id = rental.customer_id
GROUP BY customer.first_name, customer.last_name
ORDER BY unique_rentals DESC
```

Python Pandas:

python

Copy code

```
query = """
SELECT customer.first_name, customer.last_name, COUNT(DISTINCT rental.inventory_id) AS unique_rentals
FROM customer
JOIN rental ON customer.customer_id = rental.customer_id
GROUP BY customer.first_name, customer.last_name
ORDER BY unique_rentals DESC
LIMIT 1;
"""

df = pd.read_sql_query(query, conn)
print(df)
```

Soru 97: En az sayıda farklı film kiralayan müşteri hangisidir?

sql

Copy code

```
customer.first_name, customer.last_name, COUNT(DISTINCT rental.inventory_id) AS unique_rentals
FROM customer
JOIN rental ON customer.customer_id = rental.customer_id
GROUP BY customer.first_name, customer.last_name
ORDER BY unique_rentals ASC
```

Python Pandas:

python

Copy code

```
query = """
SELECT customer.first_name, customer.last_name, COUNT(DISTINCT rental.inventory_id) AS unique_rentals
FROM customer
JOIN rental ON customer.customer_id = rental.customer_id
GROUP BY customer.first_name, customer.last_name
ORDER BY unique_rentals ASC
LIMIT 1;
"""

df = pd.read_sql_query(query, conn)
print(df)
```

Soru 98: En fazla sayıda farklı müşteri tarafından kiralanan film hangisidir?

SQL:

sql

Copy code

```
SELECT film.title, COUNT(DISTINCT rental.customer_id) AS unique_customers
FROM film
JOIN inventory ON film.film_id = inventory.film_id
JOIN rental ON inventory.inventory_id = rental.inventory_id
GROUP BY film.title
ORDER BY unique_customers DESC
LIMIT 1;
```

Python Pandas:

python

Copy code

```
query = """
SELECT film.title, COUNT(DISTINCT rental.customer_id) AS unique_customers
FROM film
JOIN inventory ON film.film_id = inventory.film_id
JOIN rental ON inventory.inventory_id = rental.inventory_id
GROUP BY film.title
ORDER BY unique_customers DESC
LIMIT 1;
"""

df = pd.read_sql_query(query, conn)
print(df)
```

Soru 99: En az sayıda farklı müşteri tarafından kiralanan film hangisidir?

sql

 Copy code

```
SELECT film.title, COUNT(DISTINCT rental.customer_id) AS unique_customers
FROM film
JOIN inventory ON film.film_id = inventory.film_id
JOIN rental ON inventory.inventory_id = rental.inventory_id
GROUP BY film.title
ORDER BY unique_customers ASC
LIMIT 1;
```

Python Pandas:

python

 Copy code

```
query = """
SELECT film.title, COUNT(DISTINCT rental.customer_id) AS unique_customers
FROM film
JOIN inventory ON film.film_id = inventory.film_id
JOIN rental ON inventory.inventory_id = rental.inventory_id
GROUP BY film.title
ORDER BY unique_customers ASC
LIMIT 1;
"""

df = pd.read_sql_query(query, conn)
print(df)
```



Soru 100: Hangi mağazada, hangi kategoride en fazla kazanç sağlanmış?

```
SELECT store.store_id, category.name AS genre, SUM(payment.amount) AS total_revenue
FROM store
JOIN staff ON store.store_id = staff.store_id
JOIN rental ON staff.staff_id = rental.staff_id
JOIN inventory ON rental.inventory_id = inventory.inventory_id
JOIN film ON inventory.film_id = film.film_id
JOIN film_category ON film.film_id = film_category.film_id
JOIN category ON film_category.category_id = category.category_id
JOIN payment ON rental.rental_id = payment.rental_id
GROUP BY store.store_id, genre
ORDER BY total_revenue DESC
LIMIT 1;
```

Python Pandas:

python

 Copy code

```
query = """
SELECT store.store_id, category.name AS genre, SUM(payment.amount) AS total_revenue
FROM store
JOIN staff ON store.store_id = staff.store_id
JOIN rental ON staff.staff_id = rental.staff_id
JOIN inventory ON rental.inventory_id = inventory.inventory_id
JOIN film ON inventory.film_id = film.film_id
JOIN film_category ON film.film_id = film_category.film_id
JOIN category ON film_category.category_id = category.category_id
JOIN payment ON rental.rental_id = payment.rental_id
GROUP BY store.store_id, genre
ORDER BY total_revenue DESC
LIMIT 1;
"""

df = pd.read_sql_query(query, conn)
print(df)
```